

“Is Stephen Curry really a guard? New perspective on players typology using functional data analysis”: Reviewer Responses

Steven Golovkine and Edward Gunning.

GENERAL/ OVERALL RESPONSES

We thank the associate editor and the two reviewers for their thorough and detailed review of our manuscript.

ASSOCIATE EDITOR

COMMENT # 0.1

The paper demonstrates the use of functional data analysis methods applied to shot chart densities in basketball for investigating common patterns among players. The shot chart is an important object of quantitative analysis in basketball, particularly when investigating spatial features in offensive strategies and performance. While the referees are in agreement that the paper targets a problem of interest, their reports contains several suggestions for revision. Below are some key issues with the paper, some of which overlap with comments from the two referees, that should be carefully and fully addressed in a revision, in addition to other points raised by the referees.

Reply:

LEAVE UNTIL LAST.

COMMENT # 0.2

In the introduction, several methods for analyzing shot charts are summarized, including several based on intensity surfaces. The authors cite the requirement of a discretized grid for these methods as the reason for pursuing a density-based analysis, but this is not convincing. All functional data analyses must, in the end, be performed on a discrete grid, and it is not obvious that the estimated (log) intensity surfaces could not also be used as functional data objects. Another key difference between bivariate densities and intensity surfaces is that the latter contain information about shot volume, an aspect completely absent in the density representation. Please provide a more thorough justification for using densities as data objects as opposed to existing approaches.

Reply:

We acknowledge this point, so here we expand on why we chose density surfaces over intensity surfaces:

- **Intensity surfaces:** mean number of shots expected per unit area. These are *not* normalized. Players who play more games, or just shoot more in general can have different intensities despite having same underlying patterns^a.
- **Density surfaces:** These are normalized (densities integrate to 1), hence more suitable for comparing shot profiles, especially across longer time periods. The only drawback is that they can lack information shot volumes.

Therefore, purely from a behaviour characterization standpoint, we prefer densities because they are normalized and hence allow to make fair comparisons of shooting *patterns* that are not affected by, e.g., number of games or minutes played. However, we acknowledge that an overall shot volume metric, normalized where appropriate, e.g., total shots per game, could lead to an improved characterization in future work. We have added the following sentence to the introduction to reflect this:

“Densities are computed separately for each shot type (made and missed) and each player and are therefore normalized for each individuals total number of each shots. We prefer this over an un-normalized representation that contains information about overall shot volume, so that we can make make fair comparisons of shooting patterns that are not affected by, e.g., number of games or minutes played.”

We have also added the following sentences to the discussion:

“Secondly, this work focused on characterising the spatial patterns of made or missed shots. The normalization in the density estimation step, performed separately for the made and missed shots, inherently removes information about total shot volume or shooting efficiency (e.g., percentage chance of success). Future work could include an overall measure of shot volume or shooting efficiency to improve the characterization of players’ behaviours.”

Additionally, the point on FDA methods often requiring discretizations (on a fine grid) is sometimes correct. However, when a basis representation is used, the calculation is performed using the basis coefficients rather than the discretized objects (see, e.g., Ramsay and Silverman, 2005). We argue, however, that there is not a clear consensus from the cited works on how to discretize a (half) basketball court:

1. Miller et al., 2014 and Franks et al., 2015 use one-square-foot tiles for “computational tractability of LGCP [(log Gaussian Cox process)] inference” and they “expect this tile size to capture all interesting spatial variation”.
2. Yin et al., 2022 also use one-square foot tiles (“This rectangle was divided into $50 \times 35 = 1750$ equally-sized grid boxes of $1\text{ft} \times 1\text{ft}$ ”).
3. Reich et al., 2006 divided the court into an 11×11 grid based on distance and angle.
4. Hu et al., 2023 partitioned the court into $1\text{ft} \times 2\text{ft}$ blocks.

5. Fu and Stasko, 2024 divided the court “into a series of contiguous smaller regions for aggregation and convenience”.
6. Hu et al., 2021 partitioned the court to disjoint triangles and used this mesh of discrete sampling locations to estimate a continuous surface in space via interpolation.
7. Jiao et al., 2021 partitioned the court into 50x35 grid boxes of 1ft by 1ft.
8. Wong-Toi et al., 2022 divided the front court into twelve zones.
9. Yin et al., 2023 discretized the court to generate a 25×18 matrix as the main variable of interest.

While discretization is necessary for these approaches and the justification for each type can be understood individually, we argue that the continuous FDA approach removes the subjectivity that may arise.

^aThough, some approaches employ a post-hoc normalization step, see step 3 of Miller et al., 2014.

COMMENT # 0.3

While it is true that early work in functional data analysis did include applications of FPCA (or other standard FDA methods) directly to densities (Kneip & Utikal, 2001; Nerini & Ghattas, 2007; Ramsay & Silverman, 2005), these methods have been superseded in the last decade by non-linear dimension reduction methods due to the fact that density properties are not retained by linear decompositions (Bigot et al., 2017; Delicado, 2011; Petersen & Müller, 2016). Many if not all of these newer methods can only be applied to univariate densities; however, the authors should include a discussion about the potential pitfalls of FPCA when applied to functional data that, by nature, satisfy nonlinear constraints (positivity and integration to 1). If possible, some attempt to mitigate these issues for the bivariate shot chart densities should be made.

Reply:

We acknowledge this point – we have applied FPCA, a conventional linear dimension reduction technique for standard functional data, i.e., curves assumed to belong to the space of $\mathcal{L}_2(\mathcal{T})$ of square-integrable functions on $\mathcal{T}$. However, densities belong to a constrained space – they are non-negative and integrate to 1. Therefore, reconstructions given by standard FPCA are not guaranteed to satisfy density constraints. As such, alternative non-linear approaches have been proposed for modeling univariate densities, e.g., transformations that map the densities to an unconstrained

space before doing FPCA (Petersen & Müller, 2016). However, we have chosen to apply standard FPCA, acknowledging its limitations, because:

1. Conventional FPCA is tractable, it is straightforward to describe to a general data-science audience. We can interpret the individual FPCs as a shift of the mean density (Kneip & Utikal, 2001). Furthermore, standard computational FPCA methods can be employed to deal with the high-dimensionality of the two-dimensional surfaces.
2. Although specialized geometrically-aware density approaches can be can give better performance, standard FPCA has been shown to provide a reasonable approximation in many cases (Petersen & Müller, 2016).
3. If FPCA approximations of individual densities take negative values, they can be projected back into the density space, by retaining only the positive part and re-normalizing (Dai & Müller, 2018; Petersen & Müller, 2016).

Therefore, we decided to use the standard FPCA approximation rather than develop a new method by extending existing univariate density approaches to the bivariate case. We do, however, now acknowledge this as a limitation in our discussion:

“We employed standard FPCA which assumes that observations are realisations from an unconstrained space of functions. However, densities are constrained because they must be non-negative and integrate to 1. Reconstructions from standard FPCA do not necessarily satisfy the non-negativity constraint so specialized methods have been developed (Petersen & Müller, 2016). However, these are for the univariate case. Rather than develop or extend these methodological approaches to the bivariate case, we apply the more tractable, standard FPCA approach while acknowledging its limitations. If required, reconstructions that lie outside the space of densities can be projected back into the density space.”

COMMENT # 0.4

The choice of k-medoid clustering is also not discussed in detail. As suggested by the reviewers, please investigate the robustness or variability of your findings to more clustering methods, or provide a justification for this particular approach.

Reply:

We will address all comments regarding the clustering method here. According to Jacques and Preda, 2014, there are mainly four possibilities to cluster functional data: raw-data clustering, 2-step methods, non-parametric clustering and model-

based clustering.

- Raw-data clustering consists in directly clustering the discretization of the surfaces at some points. Because we discretized the density on a grid of 201×201 points (that would be flattened to vectors of 40401 points), this is not ideal to cluster the data.
- 2-step methods consist of, first, reducing the dimension of the data, usually by expanding the data into a basis, and second, to perform the clustering on the basis coefficients.
- non-parametric clustering consists in using specific distances between surfaces combined with classic clustering algorithms.
- model-based clustering assumes probability distributions on the underlying data. As the notion of probability density function is not well defined for functional data, the probability distributions are usually assumed on the coefficients that describe the data.

According to Zhang and Parnell, [2023](#), clustering methods are split into 2 categories: vector space and function space. (Not sure that we should use this paper.)

COMMENT # 0.5

A key drawback of the analysis is its lack of assessment of sampling variability. This is critical both for the reliability of the interpretation of the functional principal components (particularly those with smaller eigenvalues) as well as the clustering results. Without some inferential instrument (e.g., confidence bands), it is impossible to judge the importance of the findings.

Reply:

We acknowledge this as a limitation, though it is slightly less relevant than in typical scientific analyses. We computed these components based on the whole population of NBA players over a specific period, rather than a sample. Therefore, we are not necessarily trying to make inferences about the modes of variability in a larger population. That being said, we acknowledge that a population does not have to exist physically – e.g., the hypothetical notion of a new NBA player not observed or from another time period.

Therefore, as a simple sensitivity check, we have performed the following exercise and added it to the appendix:

- Construct 5 bootstrap samples.
- For each bootstrap sample, compute the MFPCA.
- Display the mean and the first 4 MFPCs from each bootstrap sample.

The shapes of the mean function and the first two principal components are very consistent. There is slight sample-to-sample variability for the third component. This result is intuitive – the mean and leading components, explaining the most variability, represent the most common modes of variation in the data and hence are more stable from sample to sample. Overall, this simple sensitivity check indicates that the prominent modes of variation are stable.

COMMENT # 0.6

Although the analytical steps are clearly laid out, it is not obvious what the clustering should reflect relative to player positions. Based on the title and some of the conclusions stated by the authors, the underlying assumption seems to be that there is something unexpected or interesting about the lack of agreement between the clustering and designated player positions. However, there seems no obvious reason why these should coincide given different coaching styles and combination of player strengths. Please provide more concrete guidance about how the proposed analytical pipeline can be useful.

Reply:

To address this, we have added the following description and caution to the methods section:

“Moreover, we will compare the estimated clusters to the $k = 5$ nominal positions provided by the NBA. We do not expect perfect agreement – defence, play-making and movement all play a role in addition to shooting. However, because shooting behaviour is a key component of a player’s overall game, the positions comparison can help inform the labels we provide to our clusters.”

COMMENT # 0.7

Please carefully check the interpretation of each of the functional principal components. For instance, I do not believe it is the case that the first component reflects players who shoot more or less than average. This is because the functions being analyzed contain no information about overall shot volume, but only the distribution of shots. I also do not believe it is possible that the entire first component is positive, since it is known – at least in the univariate case

(Kneip & Utikal, 2001) – that principal components for densities must integrate to zero.

Reply:

Thank you for this comment. Indeed, this interpretation was initially phrased incorrectly. We have now replaced this explanation with a correct interpretation, in terms of relative numbers of shots missed and shots made, rather than overall shooting volume. Regarding the negativity-positivity comment, we failed to interpret/ describe the small blue (negative) dot directly underneath the basket for the missed shots. An interpretation has been suitably added.

COMMENT # 0.8

In addition to potentially comparing with additional clustering methods, further comparisons should be made between the proposed density-based approach and existing intensity-based approaches (functional or non-functional) in order to highlight advantages, if any, of the proposed pipeline.

Reply:

See Comment 1.4 for a discussion on the different clustering approaches.

REVIEWER # 1

COMMENT # 1.1

The current research is characterizing NBA player shooting behavior using functional data analysis. The authors treat made and missed shot locations as bivariate functional data on the offensive half-court and perform a multivariate functional principal component analysis to extract dominant shooting patterns. Then, using k-medoids clustering on the resulting player-specific scores, they generate a data-driven typology of player shooting styles, questioning the validity of traditional NBA positional labels. Several conceptual and technical issues limit both the novelty and the inferential credibility of the results. I therefore recommend a revision before the paper can be considered for publication.

Reply:

Leave until end.

COMMENT # 1.2

While the paper is intended for a quantitative audience, some statistical steps, especially the MFPCA, are described at a relatively informal level. There is no discussion of convergence or uncertainty in the functional principal components.

Reply:

Given the applied nature of the paper, we have not presented convergence results for MFPCA. However, a full theoretical treatment of the method can be found in Happ and Greven, 2018. While we are primarily concerned with describing the variation in an entire population of NBA players, rather than making inferences using a sample, uncertainty quantification for the MFPCs is not of primary concern either. However, to assess stability / sensitivity of our approach, we added a simple bootstrap step (see comment 0.6).

COMMENT # 1.3

The authors do not evaluate the stability of their clustering. Furthermore, they do not provide external validation (e.g., prediction, test–train splits, comparison with existing clustering approaches). Please consider using metrics such as Adjusted Rand Index or Silhouette scores to assess cluster quality. Alternatively, compare results with simpler clustering methods (e.g., k-means on shooting frequency by location bins) to demonstrate added value.

Reply:

Adjusted Rand Index (ARI) and Silhouette coefficients have been computed. ARI between the NBA clusters and the proposed clusters are close to 0 but this is expected as we are looking for another typology than usual NBA players. The Silhouette coefficient for the k-medoids with equal weights is equal to 0.04, while the Silhouette coefficient for the k-medoids with variance weights is equal to 0.4. We also computed the Silhouette coefficient of the NBA clustering using the distance matrix used for the k-medoids with equal weights (-0.05) and using the distance matrix used for the k-medoids with variance weights (-0.1).

COMMENT # 1.4

While contrasting the clusters with NBA-defined positions is informative, the resulting disagreement is not deeply analyzed. Are there specific cases where agreement is worse?

Reply:

We do highlight that there is agreement for the center category:

“The only exception is the NBA-defined ‘center’ category, which aligns with the cluster 1 in the k-medoids results when weighted by the variance explained.”

Since the majority of both clusters and positions (each axis of the confusion matrix) disagree, it is difficult to pinpoint the most disagreement. **Steven: Were you going to add something about guard split into two cluster?**

COMMENT # 1.5

The paper uses a 2D KDE with Gaussian kernel and Silverman’s rule but does not specify whether the bandwidth is optimized globally, per player, or per component.

Reply:

A different bandwidth is considered for each component (missed and made shots chart) of each player. We have added a sentence to the main text to clarify this.

COMMENT # 1.6

The model jointly treats made and missed shots, but no attention is given to efficiency metrics (i.e., shooting percentage), which are arguably central to evaluating shot quality.

Reply:

We have now added the following to the discussion to acknowledge this point:

“Secondly, this work focused on characterising the spatial patterns of made or missed shots. The normalization in the density estimation step, performed separately for the made and missed shots, inherently removes information about total shot volume or shooting efficiency (e.g., percentage chance of success). Future work could include an overall measure of shot volume or shooting efficiency to improve the characterization of players’ behaviours.”

REVIEWER # 2

COMMENT # 2.1

The article is interesting and has an interesting premise but needs additional work. I have included a number of grammatical edits below. Please make these corrections and ask a peer to review the article for other edits that I may have missed. I included non-grammatical comments below the grammatical fixes. P2, L23: "his/her" not its P2, L29: "to estimate" P3, L22-23: inconsistent quotations P4, L40: characterizes P4, L50: capitalize Euclidean P5, L36: rewrite sentence L38: "short corners?" (There are only two corners and they are both short.) 41: English 42: English 43: will "shoot" Break up the first paragraph in 4.1 into multiple smaller ones.

Reply:

We thank the reviewer for their comments. We corrected the grammatical errors. Regarding the "short corners", the "short corners" refers to specific location on the court. We added a graphic that explain the different names of the court location in the Appendix.

COMMENT # 2.2

Can you explain how this is shown by the 3rd PC? "Interestingly, players who perform well near the basket also tend to attempt and miss shots from the corners"

Reply:

The explanation of the third principal components has been rewritten.

COMMENT # 2.3

Last sentence (P5, L55-56) does not make sense. What is "the axis of the hoop" or "the axis of the basket" (you use both terms)?

Reply:

The explanation of the fourth principal component has also been rewritten.

COMMENT # 2.4

Figure 5: for visualization purposes, consider normalizing the scores so that the axes all line up. Also consider adding a vertical line at $x = 0$ so you can more easily determine if the score is positive or negative. If possible, caption each graph with the interpretation you are assigning to it so that a reader can look at the graph and understand what it means for the labeled players.

Reply:

We thank the reviewer for the suggestion. We have standardized the scores and added a vertical line at $x = 0$. We also added the interpretation of the eigenfunctions that goes with the scores in the caption.

COMMENT # 2.5

P6 L51: "designated" as outliers

Reply:

We thank the reviewer for spotting the typo.

COMMENT # 2.6

P9 L51: describe the clusters in numeric order.

Reply:

The ordering of the clusters has been changed, and the clusters are now described in numeric order.

COMMENT # 2.7

P10: give a link / reference to Hawk-Eye; "player's"

Reply:

The typo (players $\rightarrow$ player's) has been fixed. The Hawkeye reference has been added.

COMMENT # 2.8

At the very least, you should report the number of guards, forwards, and centers in each cluster. (Many readers may not be able to distinguish who is a guard or who is a forward, etc, just by reading players' names.) It would also be nice if you used a quantitative metric to measure the agreement between the clusters you find and the NBA positions. If you try other functional clustering methods (as I suggest below), these comparisons would be even more interesting.

Reply:

We added the number of guards, forwards and centers in each cluster in the Appendix. We also changed the confusion matrix to show the NBA position directly. See Comment 0.4 for a discussion on metrics to assess the clustering quality.

COMMENT # 2.9

Did you consider other functional clustering approaches? It would be interesting to see if you get similar clusters (and similar levels of agreement) using a different approach.

Reply:

See Comment 0.4 for a discussion on other functional clustering approaches.

COMMENT # 2.10

Why not cluster the surfaces themselves and not the MFPCA scores using functional clustering algorithms? That would be fun to see.

Reply:

See Comment 0.4 for a discussion on other functional clustering approaches.

COMMENT # 2.11

What about just FPCA of all shot locations (as opposed to made and missed shots?) Do you gain anything from doing MFPCA in the way that you do vs just FPCA of shot locations?

Reply:

This is an interesting suggestion. Our rationale is that a players typical shooting "pattern" or "behaviour" is defined not only by where they shoot, but more specifically by where they score and miss from. In a simple example, we might consider two players who attempt a lot of shots from both the three-point line *and* under the paint. Player

A is an expert three-point shooter but weaker at closer-range shots, and vice-versa for player B. With a standard FPCA of overall shot volumes as suggested, we would characterize these player similarly. However, the MFPCA of made and missed shots would be able to decipher between the two patterns.

As noted in Comment 0.2, however, we do agree that future work could use additional information (e.g., overall shots volume, normalized or un-normalized) to provide improved characterisations.

REFERENCES

- Bigot, J., Gouet, R., Klein, T., & López, A. (2017). Geodesic PCA in the Wasserstein space by convex PCA. *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques*, 53(1), 1–26.
- Dai, X., & Müller, H.-G. (2018). Principal component analysis for functional data on Riemannian manifolds and spheres. *The Annals of Statistics*, 46(6B), 3334–3361.
- Delicado, P. (2011). Dimensionality reduction when data are density functions. *Computational Statistics & Data Analysis*, 55(1), 401–420.
- Franks, A., Miller, A., Bornn, L., & Goldsberry, K. (2015). Characterizing the Spatial Structure of Defensive Skill in Professional Basketball. *The Annals of Applied Statistics*, 9(1), 94–121.
- Fu, Y., & Stasko, J. (2024). HoopInSight: Analyzing and Comparing Basketball Shooting Performance Through Visualization. *IEEE Transactions on Visualization and Computer Graphics*, 30(1), 858–868.
- Happ, C., & Greven, S. (2018). Multivariate Functional Principal Component Analysis for Data Observed on Different (Dimensional) Domains. *Journal of the American Statistical Association*, 113(522), 649–659.
- Hu, G., Yang, H.-C., & Xue, Y. (2021). Bayesian group learning for shot selection of professional basketball players. *Stat*, 10(1), e324.
- Hu, G., Yang, H.-C., Xue, Y., & Dey, D. K. (2023). Zero-inflated Poisson model with clustered regression coefficients: Application to heterogeneity learning of field goal attempts of professional basketball players. *Canadian Journal of Statistics*, 51(1), 157–172.
- Jacques, J., & Preda, C. (2014). Functional data clustering: A survey. *Advances in Data Analysis and Classification*, 8(3), 231–255.
- Jiao, J., Hu, G., & Yan, J. (2021). A Bayesian marked spatial point processes model for basketball shot chart. *Journal of Quantitative Analysis in Sports*, 17(2), 77–90.
- Kneip, A., & Utikal, K. J. (2001). Inference for Density Families Using Functional Principal Component Analysis. *Journal of the American Statistical Association*, 96(454), 519–532.
- Miller, A., Bornn, L., Adams, R., & Goldsberry, K. (2014). Factorized point process intensities: A spatial analysis of professional. *Proceedings of the 31st International Conference on International Conference on Machine Learning - Volume 32*, I-235–I-243.
- Nerini, D., & Ghattas, B. (2007). Classifying densities using functional regression trees: Applications in oceanology. *Computational Statistics & Data Analysis*, 51(10), 4984–4993.
- Petersen, A., & Müller, H.-G. (2016). Functional data analysis for density functions by transformation to a Hilbert space. *The Annals of Statistics*, 44(1), 183–218.

- Ramsay, J. O., & Silverman, B. W. (2005). *Functional Data Analysis*. Springer.
- Reich, B. J., Hodges, J. S., Carlin, B. P., & Reich, A. M. (2006). A Spatial Analysis of Basketball Shot Chart Data. *The American Statistician*.
- Wong-Toi, E., Yang, H.-C., Shen, W., & Hu, G. (2022). A Joint Analysis for Field Goal Attempts and Percentages of Professional Basketball Players: Bayesian Nonparametric Resource. *Journal of Data Science*, 21(1), 68–86.
- Yin, F., Hu, G., & Shen, W. (2023). Analysis of Professional Basketball Field Goal Attempts via a Bayesian Matrix Clustering Approach. *Journal of Computational and Graphical Statistics*, 32(1), 49–60.
- Yin, F., Jiao, J., Yan, J., & Hu, G. (2022). Bayesian Nonparametric Learning for Point Processes with Spatial Homogeneity: A Spatial Analysis of NBA Shot Locations. *Proceedings of the 39th International Conference on Machine Learning*, 25523–25551.
- Zhang, M., & Parnell, A. (2023). Review of Clustering Methods for Functional Data. *ACM Transactions on Knowledge Discovery from Data*, 17(7), 91:1–91:34.